Technology: accelerating decisions

Modelling languages: ArchiMate (1/2)

Why model architecture?

- technical and functional interaction
- common points of reference
- clear and understandable
- multiple layers.

ArchiMate

- The Open Group modelling language for architecture.
- TOGAF: The Open Group Architecture Framework.
- Based on IEEE 1471.
- Based on Six Layers: business, application, technology, implementation/migration, strategy, physical.
- Each item has four aspects: active, behaviour, passive, motivation.

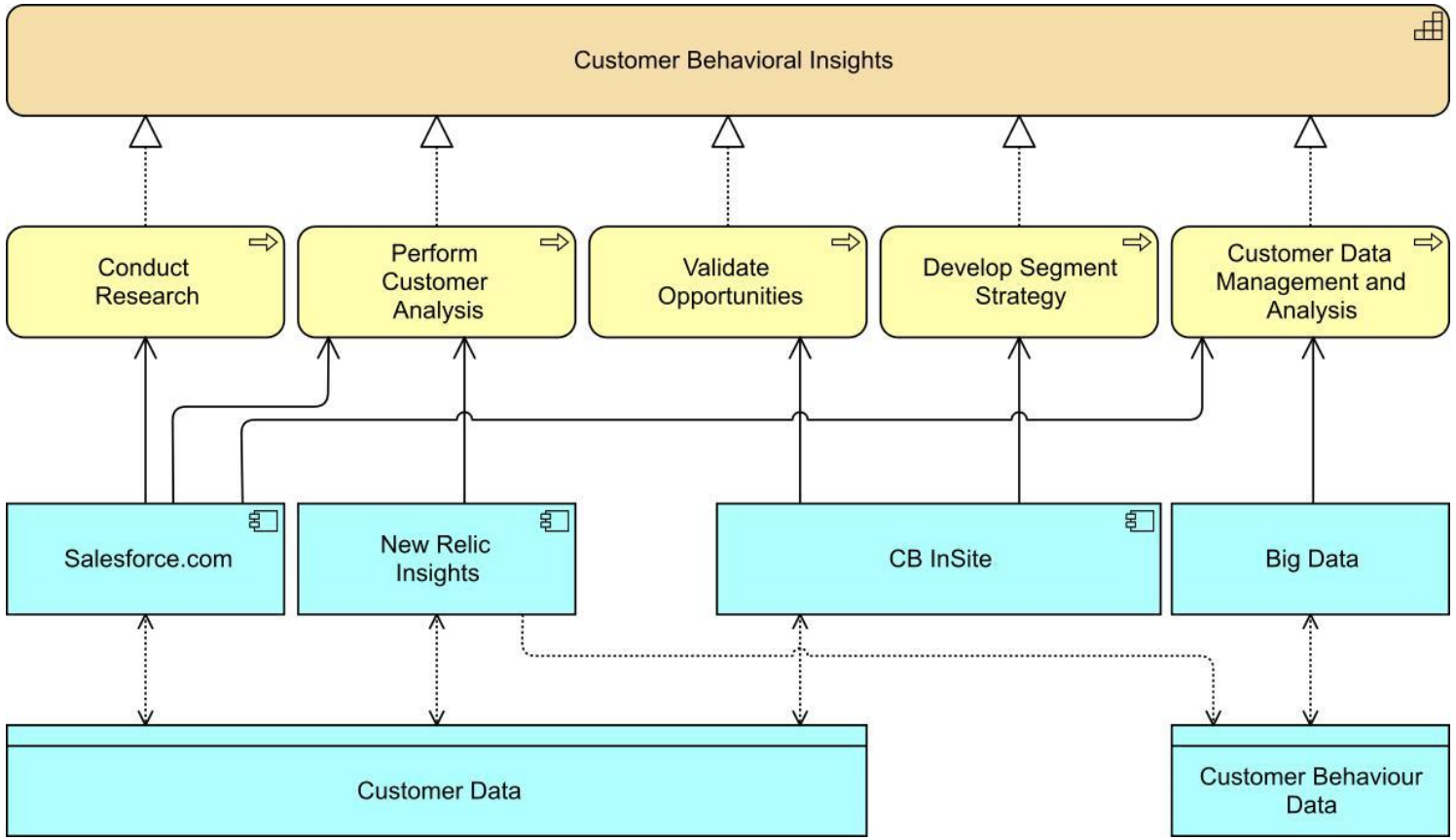
ArchiMatetool.com

- Archi as open-source modelling toolkit.



Copyright© 2025 PeopleCert International Ltd.

Modelling languages: ArchiMate (2/2)



Source: https://bizzdesign.com/blog/agile-archimate-intentional-architecture/

Copyright© 2025 PeopleCert International Ltd.

Modelling languages: UML (1/4)

Unified Modelling Language (UML)

- General purpose modelling.
- Established by Object Management Group (OMG) and published in 2005 as approved ISO standard.

Diagrams are broken into subordinate functions

- Each represents functions.
- Relate by association, inheritance, and aggregation.

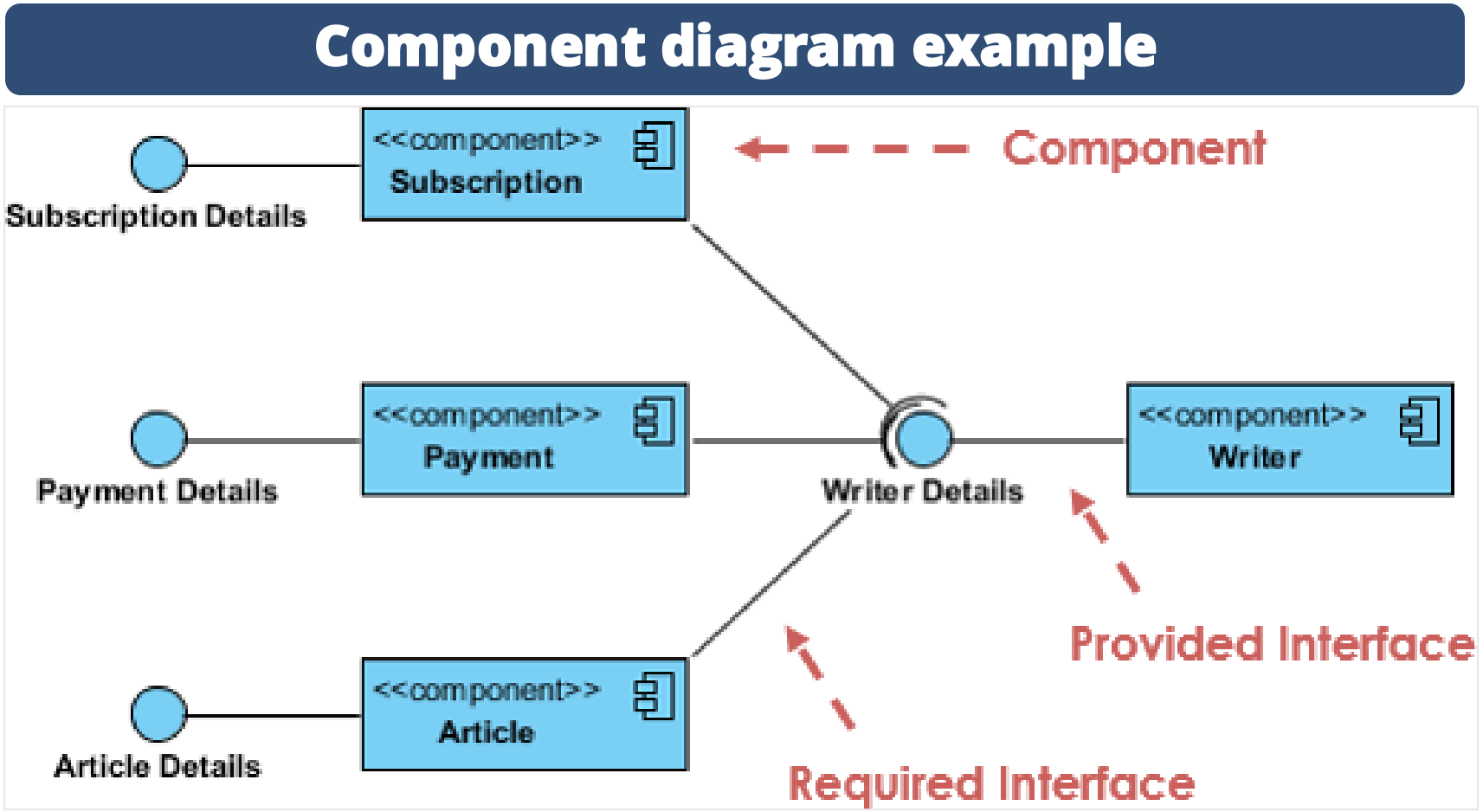
Represents multiple options Helpful in structuring software

- StarUML for open-source modelling.



Copyright© 2025 PeopleCert International Ltd.

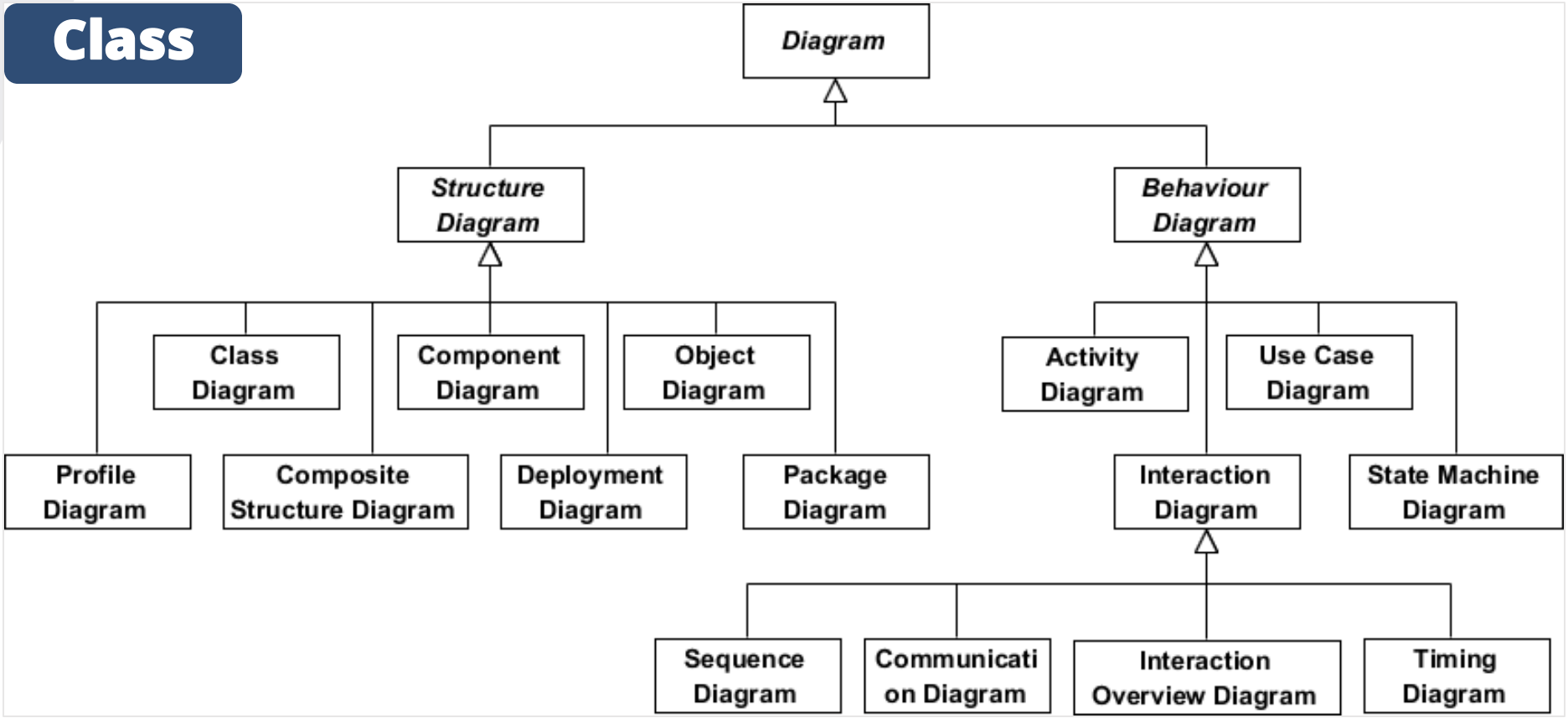
Modelling languages: UML (2/4)



Source: https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-uml/

Copyright© 2025 PeopleCert International Ltd.

Modelling languages: UML (3/4)

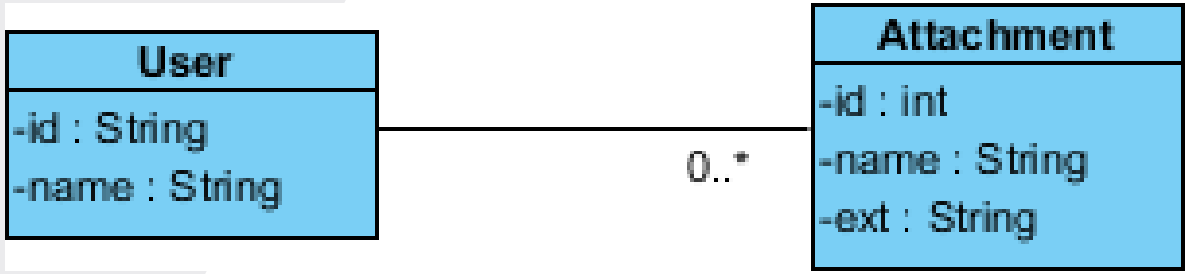


Source: https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-uml/

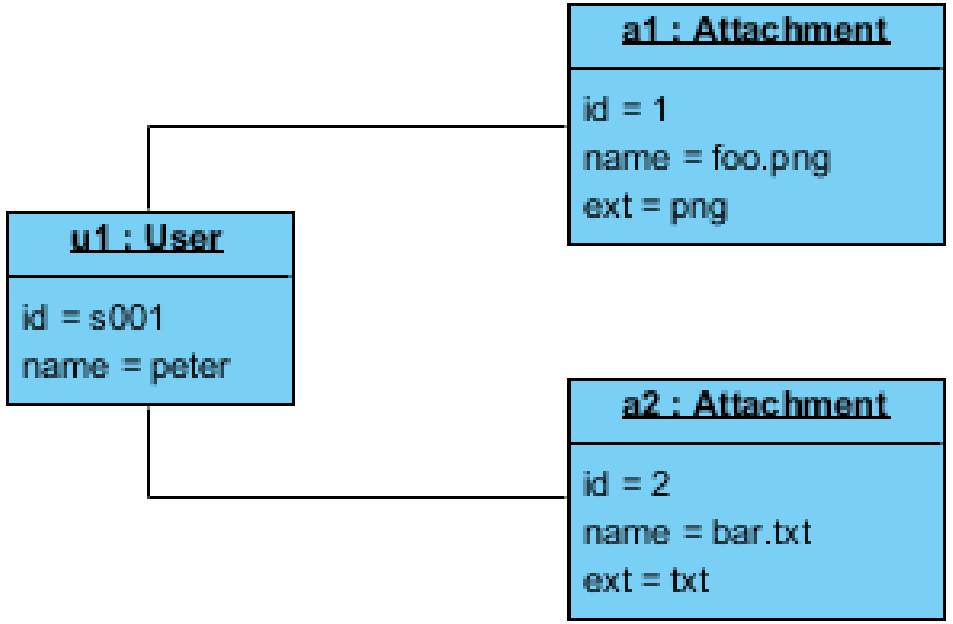
Copyright© 2025 PeopleCert International Ltd.

Modelling languages: UML (4/4)

Class diagram example



Object diagram example

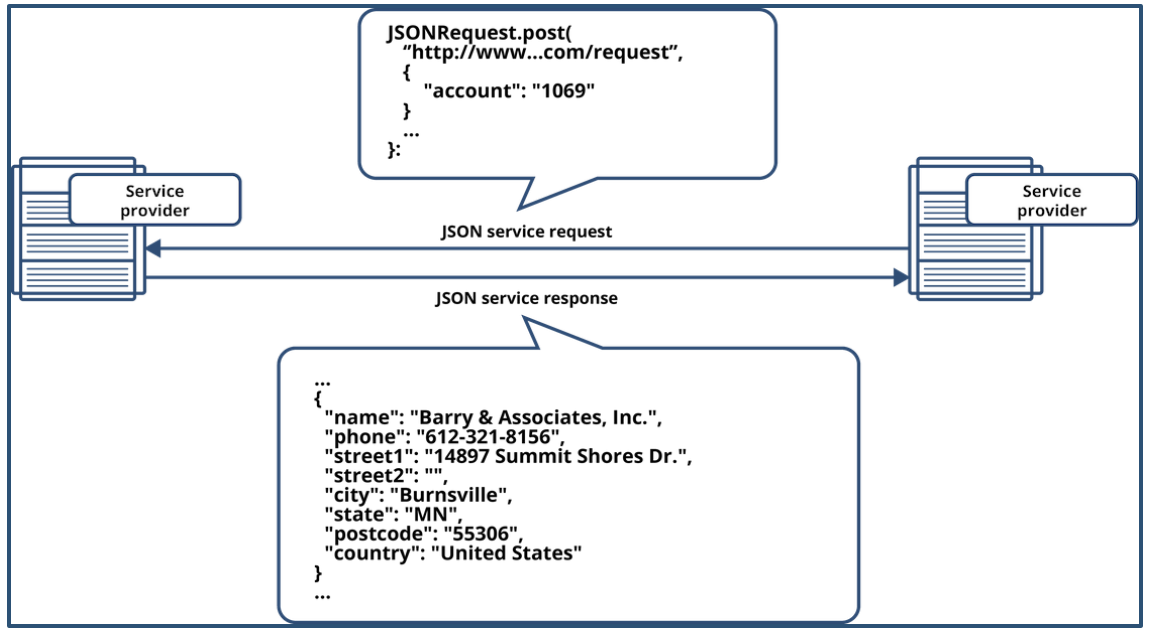


Source: https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-uml/

Copyright© 2025 PeopleCert International Ltd.

JSON

- **Java Script Object Notation (JSON)**
  - derived from the ECMAScript programming
  - language standard
  - text based format
  - represents structure
  - used to transmit data for web applications.
- **Allows communication between application and architecture.**
- **Fundamental to DevOps pipelines.**



JSON request converting text to object

```
request.open('GET', requestURL);
request.responseType = 'text'; // now we're getting a string!
request.send();

request.onload = function() {
  const superHeroesText = request.response; // get the string
from the response
  const superHeroes = JSON.parse(superHeroesText); // convert
it to an object
  populateHeader(superHeroes);
  showHeroes(superHeroes);
}
```


Copyright© 2025 PeopleCert International Ltd.

Node.js



Node.js

- open-source, cross-platform, backend, Java
- environment
- supports event-driven architecture
- capable of asynchronous input/output (I/O).

Primarily Java but can compile from other languages

- uses .npm to load packages
- creates scalable scripts without threading.

Prevents architecture from locking based on multiple requests.

Copyright© 2025 PeopleCert International Ltd.

Cloud architects (1/2)

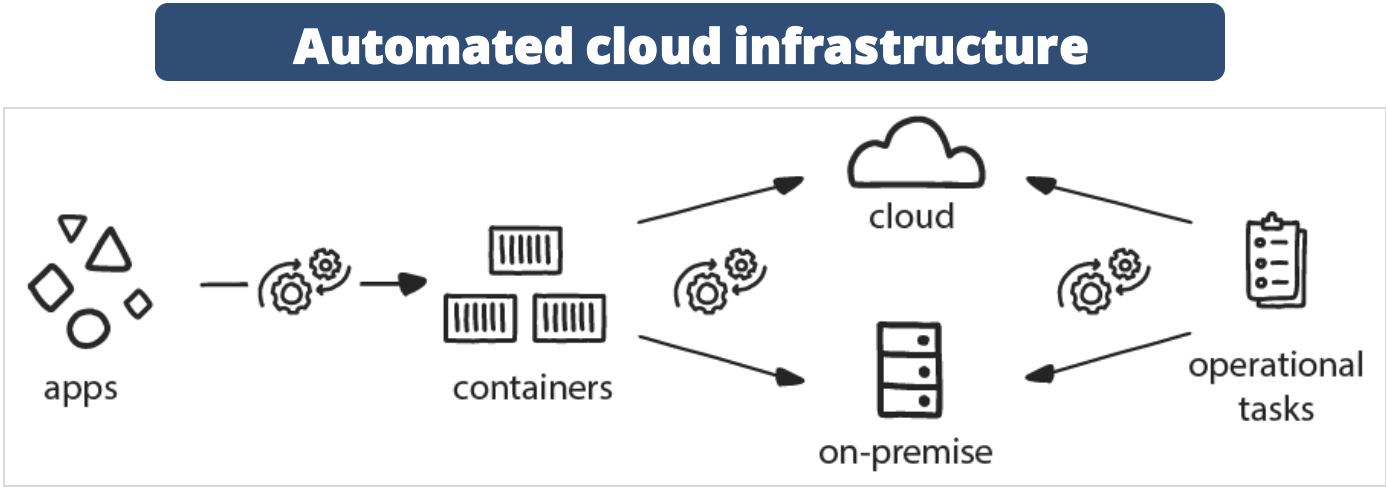
Which part of the cloud?

Based on use case and ATAM

- IaaS
- PaaS
- Software as a Service (SaaS)
- Container as a Service (CaaS)
- ‘your label here’ as a service.

Cloud-native principles

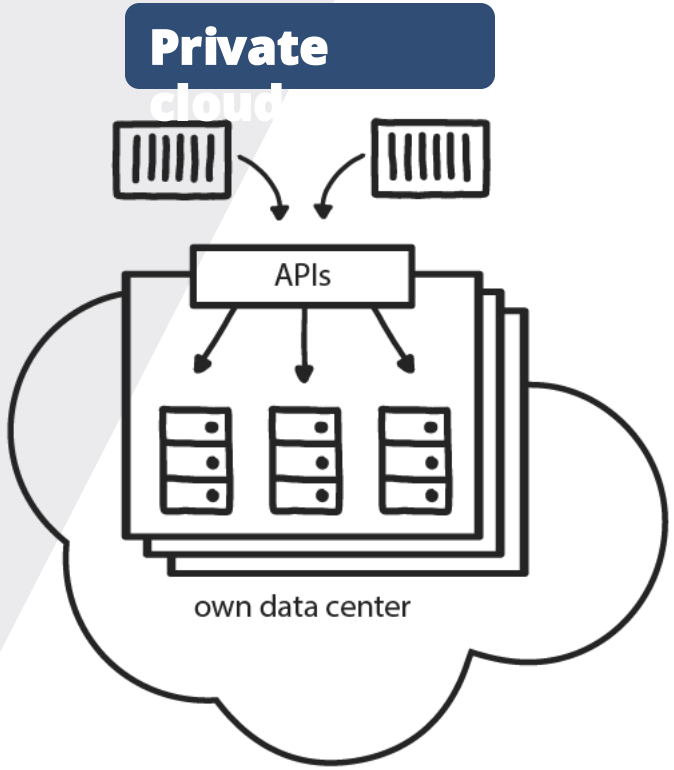
- containerization
- dynamic management
- microservices
- automation
- orchestration.



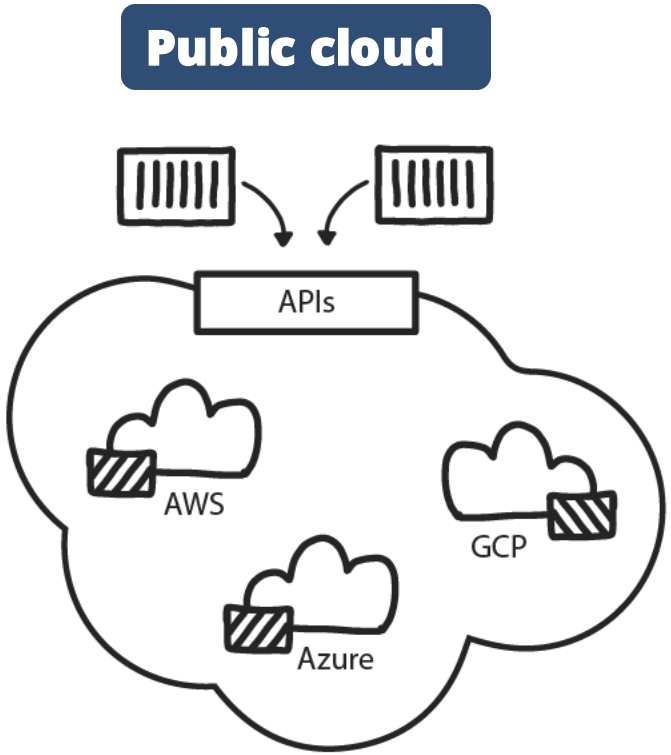
Source: https://www.cnpatterns.org/infrastructure-cloud/private-cloud

Copyright© 2025 PeopleCert International Ltd.

Cloud architects (2/2)



Source: https://www.cnpatterns.org/infrastructure-cloud/private-cloud



Source: https://www.cnpatterns.org/infrastructure-cloud/public-cloud



Copyright© 2025 PeopleCert International Ltd.